# Prime Factorization — Homework

*Lesson 6-13 • Standard 6.NOS.4*

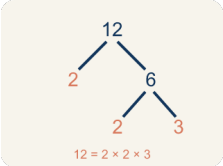
**Name:** _____ **Period:** _____ **Date:** _____

## Today's Goals

Content: I can write a number as a product of its prime factors using a factor tree.

Language: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

## Key Words



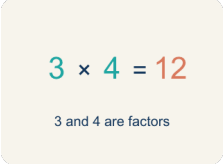

**Prime Factorization (Factorización prima)** — Writing a whole number as a product of only prime numbers.



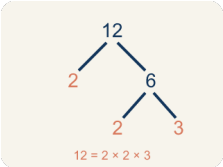

**Prime number (Número primo)** — A number bigger than 1 that you can only divide by 1 and itself.




**Composite number (Número compuesto)** — A number bigger than 1 that you can divide by more than just 1 and itself.




**Factor (Factor)** — A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because 12 ÷ 3 = 4.




**Factor tree (Árbol de factores)** — A picture that splits a number into its prime numbers, step by step.




**Exponent (Exponente)** — A small number that tells how many times to multiply a number by itself.

## Practice

*Show your work. Use the blank space under each problem.*

**1. Which of the following is a prime number?**

A. 9

B. 15
C. 17
D. 21

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**2. What is the prime factorization of 30?**

A. 2 × 3 × 5
B. 2 × 15
C. 3 × 10
D. 5 × 6

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**3. Match each number to its prime factorization.**

| Item | Answer choices |
|---|---|
| ____ 12 | F. 2 × 5 × 5 |
| ____ 18 | E. 3 × 3 × 5 |
| ____ 20 | D. 2 × 2 × 7 |
| ____ 28 | C. 2 × 2 × 5 |
| ____ 45 | B. 2 × 3 × 3 |
| ____ 50 | A. 2 × 2 × 3 |

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**4. Build the factor tree. Write each number as a product of primes.**

| Number | First Split | Prime Factorization |
|---|---|---|
| 24 | 4 × 6 | |
| 36 | 4 × 9 | |
| 40 | 5 × 8 | |
| 56 | 7 × 8 | |

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**5. Sort each number: can it be expressed as a product of exactly TWO prime factors, or does it need THREE or more?**

*Groups: Product of Exactly Two Primes | Product of Three or More Primes*

- 6 → _____
- 10 → _____
- 15 → _____
- 35 → _____
- 12 → _____
- 30 → _____
- 24 → _____
- 60 → _____

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**6. Two students found different factor trees for 60. Student A started with 2 × 30. Student B started with 6 × 10. Which statement is true?**

A. Only Student A gets the correct prime factorization
B. Only Student B gets the correct prime factorization
C. They will get different prime factorizations
D. Both get the same prime factorization: 2 × 2 × 3 × 5

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**7. Spot the Common Mistake**

Step 1 (Find the prime factorization of 45, using exponents): Start: 45
Step 2 (First split): 45 = 5 × 9
Step 3 (Break down 9): 9 = 3 × 3, so 45 = 5 × 3 × 3
Step 4 (Write with exponents): 45 = $3^2 \times 5^2$

*Which step has the mistake? Explain and fix it:*

_____

_____

_____

**8. The first lock panel shows 90. What is the prime factorization of 90?**

A. 2 × 3 × 3 × 5
B. 2 × 45
C. 3 × 30
D. 9 × 10

_____

**9. The second panel needs a number whose prime factorization is $2^2 \times 3^2$. Which number unlocks it?**

A. 12
B. 18
C. 24
D. 36

_____

**10. Sort the station ID numbers: drag each one into PRIME (only 1 and itself) or COMPOSITE (more than two factors) to clear the scanner.**

*Groups: Prime Numbers | Composite Numbers*

- 19 → _____
- 37 → _____
- 41 → _____
- 63 → _____
- 44 → _____
- 99 → _____

_____